SUPPLY-CHAIN DECISION SUPPORT

SupplyChain Twin AI

A digital-twin-style workspace for mapping a supply network, exploring disruption scenarios, and understanding potential route, cost, delay, and inventory effects—with a local AI agent to help users work with the model.

Next.js + React | FastAPI + Python | Deterministic simulations | Local AI assistant

THE PROBLEM

Disruption decisions are difficult

A supplier, route, port, or fuel-cost change can affect downstream deliveries and inventory. Teams need a clearer way to explore scenarios and compare possible responses.

THE PRODUCT

Model the network before acting

Users represent facilities and transport links, run what-if scenarios, examine modeled effects, and organize follow-up mitigation actions in one workspace.

How to explain it in 3 steps

1. **Map**
   Set up suppliers, plants, ports, warehouses, customers, inventory, and routes.
2. **Simulate**
   Describe a disruption—such as a facility closure or fuel shock—and compare modeled scenarios.
3. **Respond**
   Review impacts, discuss options with the assistant, and track selected mitigation actions.

Project stage: Local-first MVP with deployment configuration prepared. A public hosted application has not yet been deployed.

A CONNECTED OPERATIONAL WORKFLOW

From network data to an actionable review

The product combines supply-network modeling, scenario analysis, supplier monitoring, and team follow-through. It is designed to make assumptions and missing data visible—not to silently fill gaps.

DIGITAL-TWIN WORKSPACE

Represent the supply network

- Create company workspaces and guide setup from supplier to customer.
- Represent suppliers, plants, ports, warehouses, distribution centers, customers, and routes.
- Review the network visually and inspect route and facility details.
- Keep synthetic demo networks clearly distinguished from company data.

WHAT-IF SIMULATION

Explore disruption scenarios

- Model facility or route closures and fuel-cost changes.
- Recalculate a feasible lower-cost route when the network allows one.
- Compare 2–4 scenarios and review modeled cost, delay, inventory, and stockout exposure.
- Use risk labels as scenario indicators, not guarantees of real-world outcomes.

SUPPLIER OPERATIONS

Monitor reported performance

- Store dated scorecards for on-time delivery, lead time, defects, and capacity utilization.
- Preview and validate CSV/Excel reports before importing valid rows.
- Compare trends, flag configured threshold breaches, and export portfolio data.
- Show missing or outdated reports explicitly; do not grade unreported suppliers.

CONTROL TOWER & COLLABORATION

Turn review into follow-through

- Track mitigation actions with owners, due dates, status, modeled impact, and actual outcome notes.
- Share workspaces using role-based owner, editor, and viewer access.
- Review workspace activity and account notifications.
- Compare recorded outcomes with modeled estimates after actions are completed.

Data quality is part of the workflow

Readiness checks
Check connectivity, route endpoints, inventory/demand inputs, cost/capacity coverage, and source references.

Transparent provenance
Mark examples as synthetic and user-submitted information as reported—not independently verified.

Review before use
Use estimates to support investigation; confirm assumptions with verified operational data.

LOCAL, CONTEXTUAL, ADVISORY

What the Supply Chain Agent does

The agent is a conversational assistant embedded in the app. It helps people understand the product and their selected workspace, interpret modeled results, and prepare network changes for human review.

A

Answer questions

Responds to questions about app workflows and supply-chain concepts. When relevant, it receives the selected company's active network context.

B

Help describe scenarios

A local model parses natural-language disruptions into structured scenario inputs. A deterministic fallback can extract supported details when the model is unavailable.

C

Explain simulation output

Summarizes route and inventory impact and suggests mitigation ideas. The language model explains results; it is instructed not to recalculate or change simulation values.

D

Draft network records

Can propose a supplier, facility, or route draft from a conversation. The user reviews the draft and explicitly saves it through the normal app flow.

Voice and privacy approach

Voice input is transcribed with a locally installed Whisper model on the backend. The recording is sent to the local backend for transcription; the design does not use browser speech recognition or send audio to a hosted transcription provider. Company reference documents are intended for local storage and are not parsed by or sent to an external service.

What the agent does not do

- It does not autonomously modify or save the company network, supplier scorecards, or mitigation actions.
- It does not independently verify uploaded or user-entered operational facts.
- It does not guarantee predictions. Scenario outputs are estimates based on the current network data and modeled assumptions.
- It is not currently part of the prepared hosted pilot: the assistant expects a privately reachable Ollama service, while the local computer's Ollama endpoint cannot be reached from Render.

> **Plain-language positioning:** AI helps users ask questions, structure scenarios, and understand results. The simulation engine calculates the modeled numbers; a person decides what to do.

IMPLEMENTATION SNAPSHOT

Technology and project story

Frontend	Next.js 16, React 19, TypeScript
Backend	Python, FastAPI, SQLAlchemy; SQLite locally and PostgreSQL support prepared for hosting
Network & simulation	NetworkX directed graph; Python route, disruption, cost, delay, and inventory calculations
AI & voice	Local Ollama language models for guidance and scenario parsing; local Whisper transcription for voice
Deployment preparation	Vercel frontend and Render API/database blueprint included; production services still need provisioning and configuration

Resume-ready project entry

SupplyChain Twin AI — Supply-Chain Decision-Support Platform
Next.js, React, TypeScript, FastAPI, Python, SQLAlchemy, NetworkX, Ollama, Whisper

- Built a full-stack supply-chain digital-twin application for mapping supplier-to-customer networks and exploring disruption scenarios.
- Implemented route and inventory impact simulations, multi-scenario comparison, supplier scorecards, configurable risk alerts, and mitigation-action tracking.
- Integrated a local AI agent for app and supply-chain Q&A, scenario parsing, result explanations, reviewable network drafts, and offline voice transcription.
- Added data-readiness checks and explicit synthetic/user-reported data labels to communicate assumptions and data limitations.

Short interview explanation

> "I built SupplyChain Twin AI, a decision-support application where a team maps its supply network, tests disruption scenarios, and reviews estimated route, delay, and inventory impacts. I added a local AI agent to help interpret scenarios and results, while keeping calculations in the simulation engine and requiring people to review any proposed changes. The current version is an MVP; reliable results depend on verified company data, and the public deployment is still pending."

Project links & honest status

Source code: github.com/mudavathsanthosh610/SupplyChain-Twin-AI

Status: The repository is public. The app is available locally for development/testing; no public production app URL has been created yet. The hosted AI assistant is excluded from the current deployment blueprint until a secure private model service is configured.

Responsible-use note: Demonstration networks contain synthetic examples. User-provided supplier and company data is not independently verified. Treat simulation and AI outputs as decision support, validate assumptions, and make operational decisions with qualified human review.